APPLYING SYSTEM DYNAMICS APPROACH TO FAST FASHION SUPPLY CHAIN: CASE STUDY OF AN SME IN INDONESIA

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ABSTRACT

Currently, many apparel companies are shifting toward the vertical integration, and accordingly, their production processes range from raw materials to ready-to-wear clothes. Fast fashion is a concept whereby retailers orientate their business strategies to reduce the lead time for fashion products to store. This is can be achieved by working on a system of in-season buying so that product ranges are consistently updated throughout the season. Since success depends on speed, in the apparel industry, fast fashion retailers must quickly respond to market demand. In addition, defining the target market is essential for running a successful fashion business. Without sufficiently analysing an appropriate setting for the target market, a company is likely to waste time, money, and resources; for instance, marketing to the wrong customers and communicating with those not interested in the products. Therefore, supply chain management has important roles in coordinating processes along the supply chain to reduce the lead time of finished goods and meet consumer demand. Small and Medium Enterprises face the challenge of learning to manage their financial performance and attain high profit without incurring redundant costs. Therefore, this research develops a model on supply chain of an SME apparel company in Indonesia and proposes a decision-support system that applies System Dynamics and helps the management identify the best business strategy.

KEYWORDS

System dynamics, fast fashion, supply chain management, SME, Indonesia

1. INTRODUCTION

Lately Indonesia has been no more than a home for original equipment manufacturers (OEM) of branded products from other countries such as Zara, Mango, Esprit, Polo, Giordano, Forever 21 and so on. Although a substantial portion of their products are made and tailored in Indonesia, ironically all the above-mentioned brands are recognized as premium or high-end labels. Fashion industry in Indonesia has basically great growth potentialities. In figure 1 the vertical axis shows style characteristics. On this graph, the position of domestic apparel (blue) brands and foreign brands (black) are plotted. Concerning style, Indonesian apparel brands are as fashionable as foreign brands, while they sell their products at more affordable price than foreign brands. Indonesian textile association (API) shows there are dozens of world-class brand of clothing manufacturers in Bandung, West Java. Indonesia actually abounds in both natural and human resources, so it can excellent apparel goods. There can be many reasons, why products of Indonesian brands are still inferior to those of foreign brands in the market. Therefore the domestic apparel firms must quickly have to identify the reasons and to improve themselves to overcome this challenge. Learning from experience of the fast fashion industries that has been successful can be a good basis for improvement.

In Indonesia, the apparel industry is included in the creative industry, and the firms of the creative industry are mostly driven by small and medium-sized enterprises (SMEs). In addition they are estimated to be able to absorb 4.9 million workers, and also SMEs have proved to be the most enduring in the face of the economic crises, both in times of at the end of the 1990s and in late 2008 and early 2009.

Until now the trend of GDP growth in the individual creative industries amounted to 2.7% for architecture; 2.4% for design; 2.6% for fashion; 5.9% for film, video and photography; 5.5% for craft; 12.5% for computer services and software; 0.6% for music; 3.9% for market and art goods; 0.2% for publishing and printing; 12% for advertising; 14.9% for interactive games; 7.2% for research and development; 6.6% for performing arts, and 6% for television and radio. Development in the creative industry in Indonesia has contributed greatly to the growth of GDP, employment, corporate activity and foreign trade. In the period of 2009-2014, the creative industries in Indonesia are expected to grow at 7-8%.

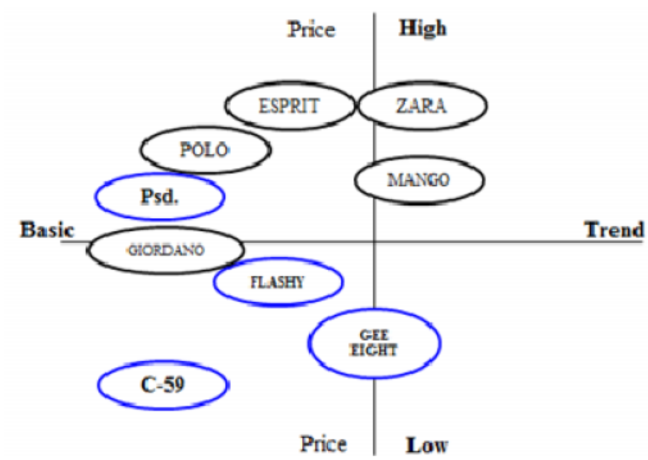

Figure 1.Domestic products position in Indonesia.

2. THE SD APPROACH

In this study, we considered a System Dynamics (SD) model to be appropriate research tools. The purpose use SD is to improving the understanding and identification of the causal relationship in the system. SD was introduced by Jay Forrester in his book, Industrial Dynamics in the early 1960s. In several areas of management research, computer simulators based on SD model are used as a means to explore the subjects' understanding and behaviour in complex situation. SD is a methodology for studying and managing complex feedback system, such as one find in business and other social systems. In fact it has been used to address practically every sort of feedback system, problem solving and policy design. The purpose of SD modelling is to improve our understanding of the ways in which an organization's performance is related to its internal structure and operating policies and then to use that understanding to design high leverage policies for success (Sterman, 2000). System dynamics is also a rigorous modelling method that enables us to build formal computer simulations of complex system and use them to design more effective policies and organizations".

3. THE CASE STUDY

The application of the proposed system is illustrated and verified through a case study. A brief description of the case company and data is given, and the proposed model is then estimated and evaluated. The case company is a typical fast fashion firm in Bandung, Indonesia that produces its own wares ranging from raw material to be ready-to-wear clothes, has three stores, a warehouse and running online sales system. This company is the founder of boudist or boutique distro community. Boutique represents the meaning of fashion for female and distro symbolize the "do it yourself" community. This company started since early 2004, has tag line for their product "Hot new and limited product everyday". The target markets of this company were women in range age are 15-30 years. This company had opened branches in Jakarta and Surabaya, but closed at the end of 2010. The company possesses data on production processes, price of product, sales, marketing strategy, product characteristic, and the number of worker in each section. Due to confidentiality, all of the actual data are concealed.

3.1. Causal Relationship

The purpose of an organization is to create more wealth for its owners. This is refers to how to create more profit. A financial measure involves some measurement of the overall profitability of the organization (McGarvey et al., 2001). Based on the goal of an organization running a business, we will illustrate a causal relationship model that provides a framework for developing model to provide decision support to run the business strategies. We analyses the result of many simulations in a fashion company from an operational point of view and from them we derive suggestions about the future business strategy in a small and medium fashion company in Indonesia. This section presents an overview of the causal relationship variables. Red arrow lines indicate the flow of material, the blue lines shows the flow of information between the factors, and the green lines indicate the financial flows in the system.

Figure 2.Causal loop diagram in an SMEs fast fashion

3.2. Dynamic Model of Customer Migration

The basic structure of frequent customer model is dynamics of migration of customers. The customer of the company comes from the total population; also coupled with the number of visitors from outside of Bandung. The targeted population is women in range age 15-29 years. This is according to company target market.

The initial of total population is set at 3,178,543 peoples and the total targeted population is 418,687. The number of visitors from outside Bandung are 150,000 peoples; the number of weekly visitors. Visit rate (**TargetRateVisit**) of 0.33, this figure is derived from the accretion in the percentage of visitor in one year (2010-2011). Considering not all consumers can wear the merchandises due to religion and culture variables, the target rate state at 0.1 from targeted population. Some of targeted population will become Sometimes Customers and moving to Frequent Customer or just stay to become sometimes customer. There are inflow (**BecomeFreq**) from sometimes customers migrate to frequent customer, and outflow (**BecomeNotFrequent**). The speed of migration actually influence by (**BecomeFreqRate**) multiply by become frequent customer rate in normaly (**BFreqNormal**); 0.002, this is a reasonable number considering that not everyone will be the buyer of this brand. The quality (**QualityEffects**) will giving affect boths on migration become frequent customer and become not frequent customer. If the quality is low, then consumers will outflow into frequent customer stock. Both the number of sometimes customer and frequent customer can be used by company to estimate demands. In this simulation there are three variables will become graphical function (**WebEffect, WomEffect, Attractiveness**) to analyze how significant the influence of word of mouth (**wom**), company website, and attaractiveness. The flow of frequent customer affected by become frequent rate and it affected by quality effects. The stock of frequent customer are affected by frequent customer rate. This rate influenced by total attractiveness that associated with merchandises. The total attractiveness here described in product attractiveness and service quality.

3.3. Dynamics Model of Attractiveness

Fetching the attention of consumers is imperative in terms of sales leverage. Regarding that, attractiveness of a brand to be the main considered in the customer's perspective. The attractiveness in this model will be state by attractiveness of product (**ProdAttractiveness**) and attractiveness of service (**Service Quality**). It has already explained in literature study the important variables in product attractiveness are design (**VarietyAttractiveness**), price (**PriceAttractivenes**s), and also quality (**QualityAttractiveness**). The variety attractiveness in this model is described into diversity (types, style, trends, colors) of clothing, price attractiveness representing how affordable price associate with the purchasing power parity of customers and value function from customer's perspective. Quality attractiveness is quality of stitching which refers to the neatness and the suitability of the pattern shape.

In service setting, higher service quality stimulates demand, but greater demand erodes service quality as waiting time increase, and accuracy, friendliness, and other experiential aspects of the service encounter deteriorate. Service quality basically affects customer impression in terms of satisfaction. It can be described on two variables, first is how attractive the interior and layout of store are designed, how the display of window store, display of new arrival cloths in the stores, how often assorting do in line displays and the ease to access the store (**ShopAttractiveness**). Base on interview and questionnaire that held by author, some evidence show that consumer visiting stores 1-3 times in a month. It shows how important presenting something new and fresh look to the store visitors. Second is staff attractiveness (**StaffAttractiveness**): can be described as the service provided by clerk in the store during the sales activity. Variable of staff attractiveness become graphical function because of intricacy to define original state. So the equation of staff attractiveness is level of service in normal state (**NormalSalesStaff**) multiply by (**SalesStaffLevel**). In this model there are seven variables that will be a graphical function, namely: VarietyAttractiveness, PriceAttractiveness, QualityAttractiveness, QualityLevel, StaffAttractiveness, ShopAttractiveness, shop access, and DisplayAttractiveness. The stock of replenishment (**DisplayLeve**l) will influence as high as for display attractiveness. The stock of Shops also affects the ShopLevel; it means if new shop established (**ROpenShops**) then the level of consumer access (**shop access**) will be higher.

3.4. Dynamics Model of Production

Basically flow of this model is fabrics inflow to production starting (**starting**) continue to Design process (drawing & cutting pattern) and moving to Sewing process (includes of accessories attach and steam process). Production starting is defined by management decision on how many item of clothing will be produce in a cycle production (**Pieces**). The items are decided by marketing team on how many articles will produce by designers. The number of items will be setup at 30 while pieces per item will become 15. The rate flow from design stock to sewing stock is presenting by SpeedDtoS. This term represent the balance of the process (**BalanceDes**) that formulated by the number of workers (**WorkersDesign**) multiplied the production capacity (**prodWD**). While the productivity in design section (**InitProdWD**). This formulation will be same applied in Sewing and Inventory stock. Finished clothing will be sent to the warehouse for Inventory and quality checked. From warehouse, attire will be distributed to the allotted stores. The new arrival item will be placed on display1. The management will hire workers as increases smoothly to its production planning and it will be affected by demand increment.

Workers will be allocated in the design (**WDesign**), sewing (**WSewing**), and quality check; including inventory (**WQuality**) also in the shops as shop keeper (**Wshops**). Market demand will affect the production plans. When demand increases, management can increase the number of workers (**HireDesigners, HireSewing, HireQC, HireShops**). This flows are bi-flow, it means when demand decreases management can reduce the number of workers. Each department is headed by a supervisor who manages production planning and adjusts to the requisite of workers then more workers will be able to done on time. On the purpose of that, workers must equip with skills to increase product quality and it can be enhanced by training. In this model, intensity of training can be set (**TrainingDes, TrainingSew, TrainingShops**) and the next chapter can be seen how much influence the training to improve company profits.

3.5. Dynamic Model of Store Operations

In store operation there are some display that will described into 3 level of display stock; display 1, display 2, and display 3. The new arrival item clothing will be display at new arrival area and for convenience will be namely display 1. The initial display1 will state at 300 pieces. These numbers represent the total clothing sending into store. After two weeks sale season activities, clothing sold to the outflow (**sold1**) and the outstanding clothing (**Disp12**) will be replaced at regularly area (**display 2**). Since the new attire will produce every 2 weeks the outstanding clothing in display 2 will be move to sale area (**display 3**). In this time the store will give discounts. In this simulation, the discount rate will state at 10% to 20%. The outflow of sold1, sold2, and sold3 will equal to the amount of sales. The remaining unsold clothing will go to Disposal stock. The initial setting for simulation are 0 for the first time, this number will changes increasingly as simulation running.

4. TESTING

Even though testing occurs constantly from the start of the modeling process, once there is a fully functional version, a severe test must take place. The test consists of various time and interval time.

4.1. Customer's Migration

Dynamic of customer's migration can be seen in figure 3. Buyers will become sometimes customers after initial purchase and they will migrate into frequent customers as attractiveness increases. There was no significant difference when the model run within 10 weeks with and without interval 5 weeks. But the graphs will quite difference in 52 weeks.

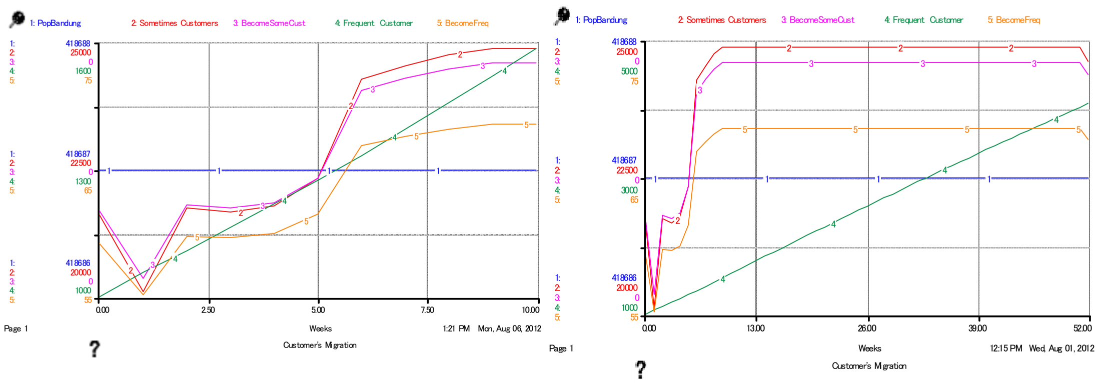

Figure 3.Customer's migration in 10 and 52 weeks

4.2. Attractiveness

Attractiveness is the most important variables in this model. Service quality from staff, product variety, and price became an attractive factor for customers. Figure 4 is a graph that defines this relationship:

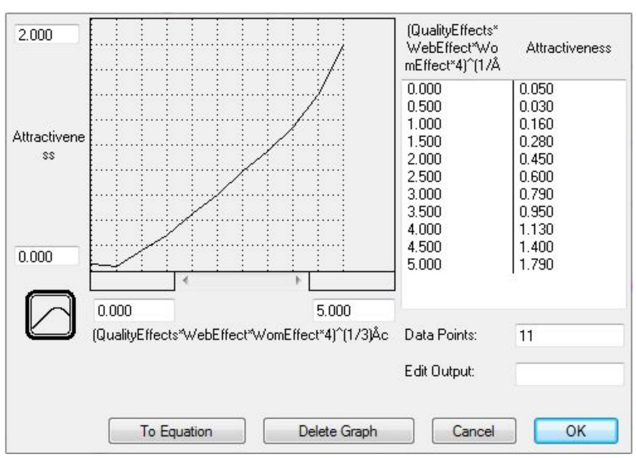

Figure 4.Relative Attractiveness

4.3. Quality level

Quality level variable is supporting by skill of designer and seamster. Quality level defined with a range of 0 to 10, intended to approximate the impact of intensify training for designer and seamster. Quality level will have a positive (>4) impact on level of workers. The shape of the curve indicates that, when the skill level of designer and seamstress high, then the effect of quality will be high too. Figure 5 is a graph that defines this relationship:

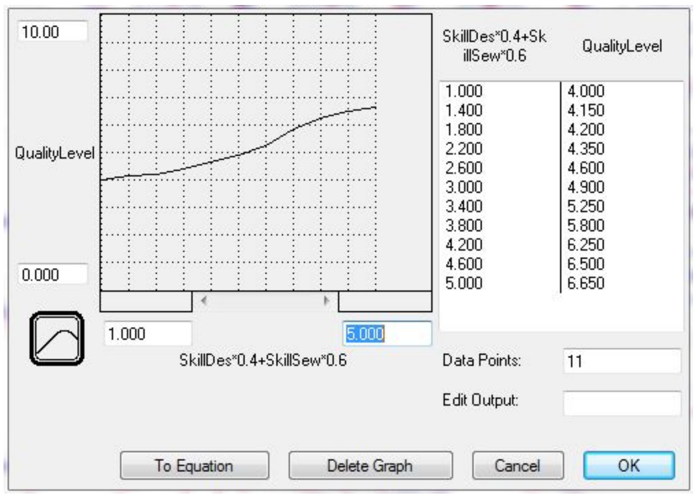

Figure 5. Quality Level

4.4. Quality Attractiveness

Quality is the most important factor to enhance attractiveness of product. This variable supposed to enrich the skill of workers. The quality attractiveness level defined with a range of 0 to 5, intended to approximate the impact of quality increases. If quality attractiveness equals quality level, then the effect is neutral (=1). Figure 6 is a graph that defines this relationship:

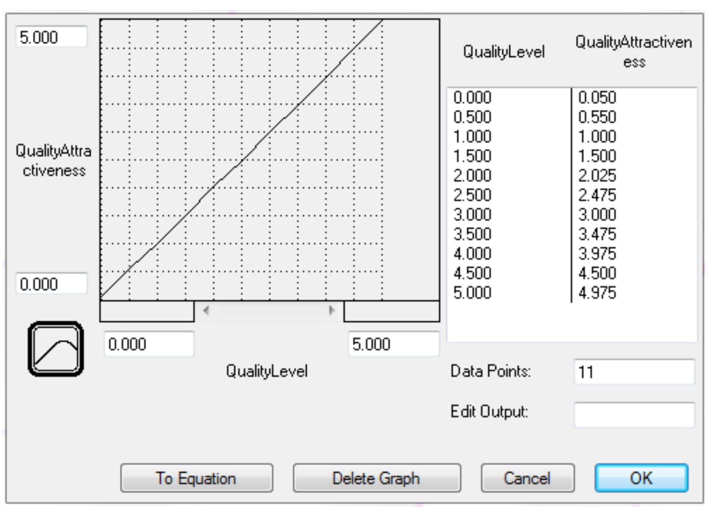

Figure 6.Quality Attractiveness

4.5. Display Attractiveness

Some evidence shows that generally customers visit their favorite outlet 1-3 times in one month. This means how important display in the store to attract visitors. Display attractiveness is representing the level of replenishment defined with a range of 0 to 5, intended to approximate the impact of display level, when compared to the normal display in the stores. Figure 7 is a graph that defines this relationship:

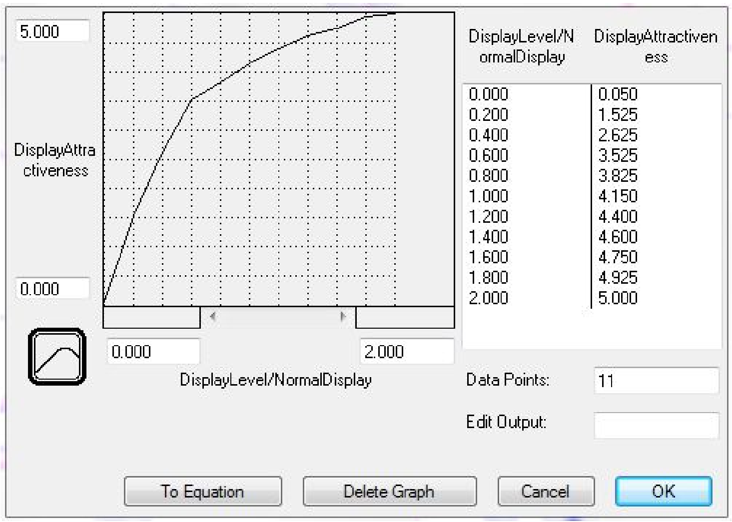

Figure 7. Display Attractiveness

4.6. Price Attractiveness

Price attractiveness is a decision variable. It enables the user to determine the desired price that is affordable to the customers to force the sales.

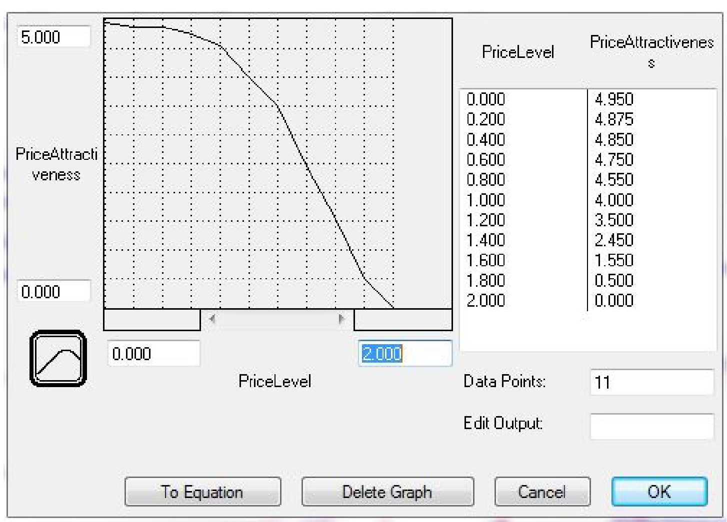

Figure 8. Price Attractiveness

However price is a sensitive variable that related to the value of customer (function and preference), therefore management should does segmentation in order to focus fulfill demand of their target market. The price variable defines with a range of 150,000 to 300,000 (Rupiah), normal price is the average of price which state at 200,000. Price attractiveness defines with a range 0 to 5, intended to approximate the impact of price level. The shape of the curve indicates that, when the price level is competitive with the value of customer, then the effect on price attractiveness will increase.

4.7. Production

In production department there are design, sewing, inventory and display1 stock. Design is representing production start and display1 is the finish garments to sell to the consumer. The following figure 9 are representing the number in each section in different time simulation.

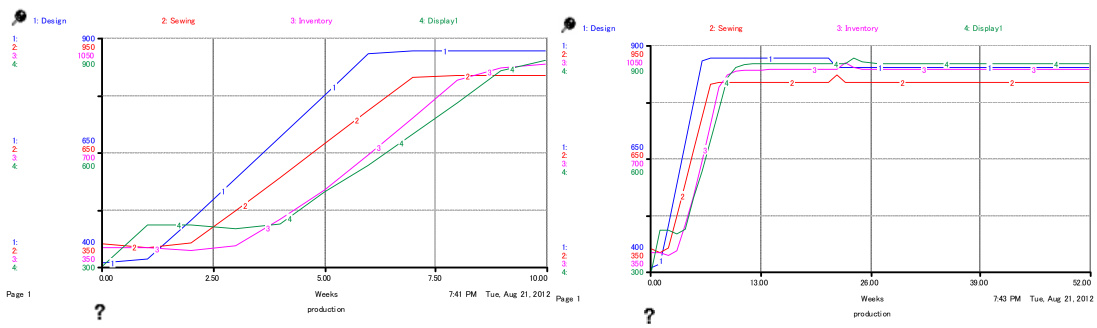

Figure 9. Production in 10 weeks and 52 weeks

The X-axis stated as the amount of garments while the Y-axis is the time simulation. The simulation was run with time interval (5 weeks) and without time interval. There are gap between design, sewing, inventory and display1. The gaps deemed as the amount of defect occur in design and sewing process. The gap between inventory and display become narrow in fifth weeks is the influence of training variable.

SpeedDtoS. SpeedDtoS is the variable that representing productivity in design section to sewing section and supporting by the skill of designer. Speed designs to sewing will high if the balance designs also high as supported by the skill of designer and seamster. The initial setting for currently productivity is 150 item/cycle manufacturing.

SkDesLevelUp. The skill designer level up is a binary variable. If the skill designer more than or equal to 2, then the value is 0, else is training designer indicating designer need to increase their skill. The following equation defines this variable:

$$\text{IF SkillDes} \geq 2 \text{ then } 0 \text{ else TrainingDes}$$

4.8. Total Display

Total display is variable that representing the aggregate of display1, display2, and display3. Table 1 is show the number of sold item.

Table 1. Sold item vs. profit in 10 weeks

Weeks	Display1	Display2	Display3	Sold1	Sold2	Sold3	Sales	Profit
0	300.00	50.00	20.00	270.00	50.00	20.00	496,500.00	9,745.64
1	405.00	30.00	0.00	364.50	30.00	0.00	587,250.00	100,465.64
2	405.00	40.50	0.00	364.50	40.50	0.00	601,425.00	114,604.64
3	395.70	40.50	0.00	356.13	40.50	0.00	588,870.00	102,006.44
4	409.47	39.57	0.00	368.52	39.57	0.00	606,204.00	119,288.60
5	492.24	40.95	0.00	437.08	40.95	0.00	710,897.65	224,374.40
6	562.79	55.16	0.00	462.29	53.07	0.00	765,080.49	278,482.59
7	645.11	100.50	2.09	448.31	81.80	1.74	784,976.49	298,289.01
8	726.99	196.80	18.70	409.58	129.86	12.61	804,816.98	318,022.01
9	811.54	317.41	66.94	367.37	168.28	36.27	821,757.35	334,833.38
Final	839.33	444.18	149.12				835,092.03	348,013.27

5 Policy Design and Evaluation

The final phase consists of using the model for policy design and evaluation. Once the model is tested and ready to use, data from company can be loaded, and the model may run under the existing policies and practices to establish the current baseline, also known as the “status quo scenario.” Then, new policies and decisions can be made, and the model is run again. Better results imply improvement, and worse results indicate another attempt. At this stage, the model becomes a great learning tool and a powerful simulator for developing entirely new strategies, structure, and decision-making rules. The first scenario works under forcing sales strategy. The second scenario is related to the increased unit demand customers. Third scenario is based on advertising and promotion in the website.

5.1. Using and Interpreting the Model

As shown in Figure 10, the model runs directly from a control panel designed for easy input of data and decision-making.

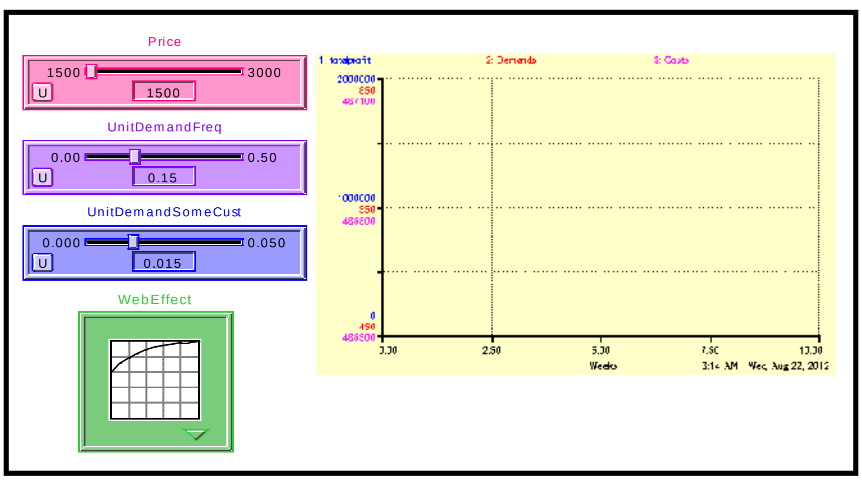

Figure 10. Control Panel

5.2. Status Quo Scenario: Baseline

In order to improve, the current baseline needs to be determined. This scenario usually not the best strategy, but is not the worst either. Table 2 shown calculated data. The result captured based on currently situation.

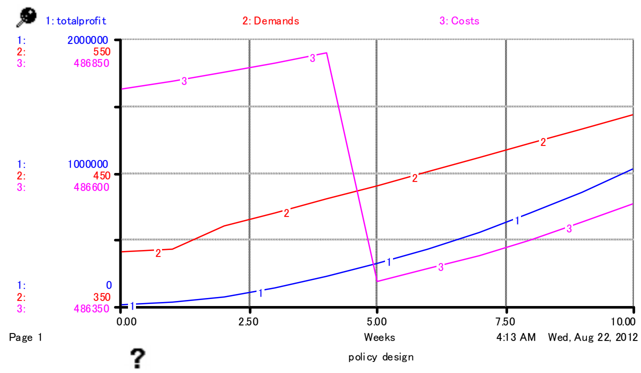

Figure 11. Status Quo Scenario

Table 2. Status Quo Scenario

Weeks	Price	Sales	totalprofit
0	1,500.00	496,500.00	0.00
1	1,500.00	530,694.15	9,745.64
2	1,500.00	556,351.78	53,670.42
3	1,500.00	570,846.16	123,236.34
4	1,500.00	584,525.91	207,278.49
5	1,500.00	598,085.10	304,980.42
6	1,500.00	613,754.32	416,673.94
7	1,500.00	628,935.41	544,012.53
8	1,500.00	643,249.10	686,505.63
9	1,500.00	657,148.98	843,283.19
Final	1,500.00	671,418.10	1,013,928.48

5.3. Aggressive Price Scenario

An alternative solution seems to be selling higher price. The aggressive price is accompanied by a 10%-20% price discount. Figure 12 shows the results achieved after running the model for 10 and 52 weeks.

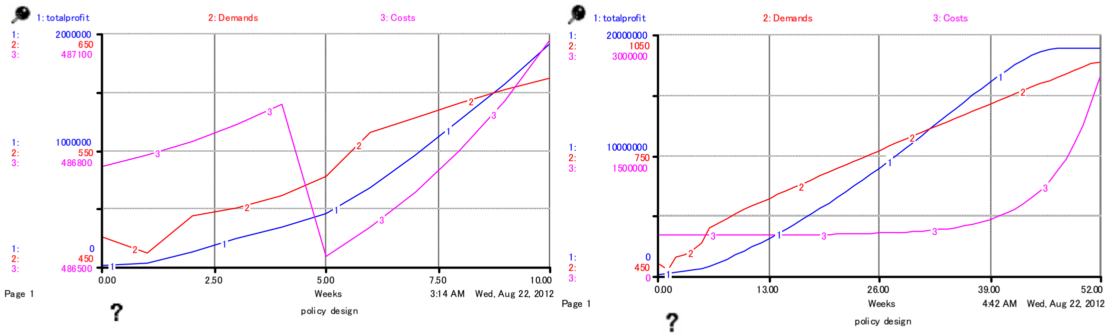

Figure 12. Aggressive price scenario-10 weeks and 52 weeks

5.4. Unit Demand Scenario

Unit demand for sometimes customer and frequent customer setting to 0.019 and 0.19. In Figure 13, profit smoothly increases but cost doing decrease and increase as training starting in fifth week.

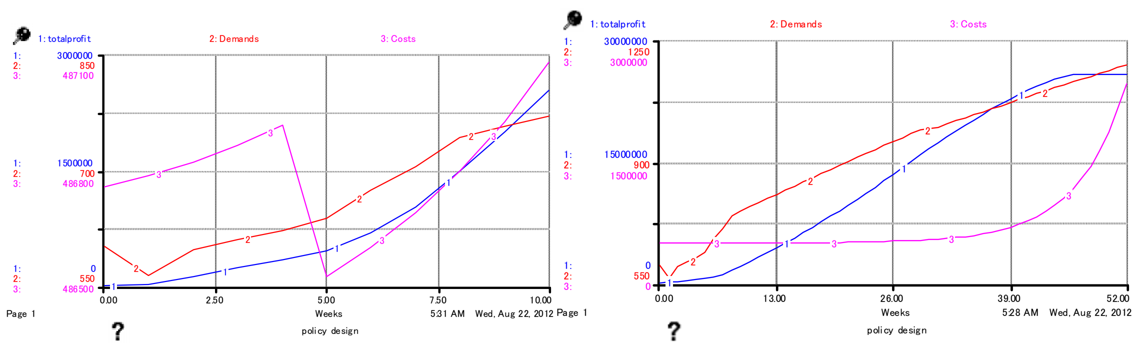

Figure 13. Unit Demand scenario-10 weeks and 52 weeks

5.5. Expand Website Scenario

Figure 14 shows that, surprisingly, after the website expands, the performance cost surge after 39th weeks.

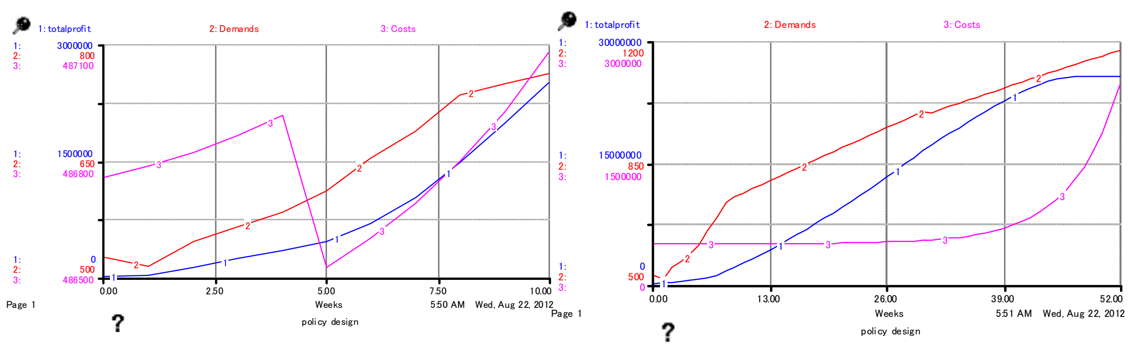

Figure 14. Website expand scenario-10 weeks and 52 weeks

6. CONCLUSIONS

In this study we constructed a decision support system, by using System Dynamics, which simulates the complexity-dynamics of fast fashion supply chain composed of small and medium enterprises (SMEs) in Indonesia. The System Dynamics model that is the core of the DSS is composed of the following sub-models concerning: Customer's purchasing behavior, Factors of attractiveness of fast fashion products, Supply chain of fast fashion goods and Store operation selling fast fashion goods.

The result of the simulations by the system suggests the following: To gain more profit the company must keep "frequent customer" and increase new customers ("sometimes customer") to purchase the merchandise. "Product attractiveness" composed of affordable price, products variety, and also "quality" greatly influences "sometimes customer" to be "frequent customer". Based on interview results the affordable price level of the target market can be deduced. Rely on this data; the company can set up the level of clothing prices.

The second factor is "service quality". "Service quality" in the model is composed of "shop attractiveness" and "staff attractiveness". "Shop attractiveness" is related to product assortment, rotation of clothing in a display area and how company define the amount of clothing with a new design for display at the new arrival area (display level). Within one month, the average customer to buy or windows shopping into store as much as 1-3 times. This is an indication that management of producing the new design and strategy of product replacements in the store are very important. It is intended that customers will find something new and fresh every time when they come into the store.

Even if these factors are met, if the availability of goods (display) in the store is low, customers will go to other stores and the store will lose the opportunity to gain sales. If sales increase, the management has resource to open a new retail store. Additional number of shop will effect on "shop level". The time taken to replenish store will become shorter if defective clothes decrease. Defective clothes can be minimized by increasing the skill of workers. With the improvement of "worker's skill" production processing time can become shorter. "Worker's skill" can be improved by training. The target for training also should be set for unskilled workers differently from skilled workers.

But these factors are trade off when the company wants to preserve the value of long-term business (customer lifetime value). In recent years, many companies have focused on how to enter markets, to meet customer's requirements to boost their market share and profit and to earn customer's loyalty in order to get greater profits in the long term. These highlight the key factors that ought to be considered in making policy decisions. In conclusion, this model provides a useful framework to understand and to analyse several factors to make decisions.

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